Chemical Equilibria and Gibbs Minimization – Analytical Gradients and Jacobians

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1. Problem statement

The single-phase Gibbs energy minimization problem can be stated as follows:

$$\begin{aligned} \min_{\mathbf{n}, n_T} \quad & \sum_{i=1}^{N} n_i \mu_i \\ \text{s.t.} \quad & \mathbf{A}^T \mathbf{n} = \mathbf{b} \\ & \mathbf{1}^T \mathbf{n} = n_T \end{aligned}$$

Here, $\mathbf{n} \in \mathbb{R}^N$ denotes the species mole vector, n_T is the total number of moles in the phase, $\mathbf{A} \in \mathbb{R}^{N \times C}$ is the elemental composition matrix, and $\mathbf{b} \in \mathbb{R}^C$ is the vector of total atoms of each conserved element introduced into the system. The quantities μ_i are the chemical potentials of the species, which we collect into the vector $\boldsymbol{\mu} \in \mathbb{R}^N$.

In other words, we seek the composition vector $\mathbf{n}$ that minimizes the total Gibbs energy while satisfying the elemental balance constraints and ensuring that the total number of moles is consistent with the phase composition.

For an ideal-gas phase, the chemical potentials are written as

$$\mu_i = \mu_i^\circ + RT \ln (x_i \gamma_i)$$

where μ_i° is the standard-state chemical potential and γ_i is the activity coefficient of species i, as given by an appropriate thermodynamic model. For gases at high temperatures, it is often reasonable to approximate $\gamma_i \approx 1$.

2. Lagrangian

This is a constrained optimization problem, which we treat by introducing the Lagrangian

$$L(\mathbf{n}, n_T, \boldsymbol{\lambda}, \eta) = \mathbf{n}^T \boldsymbol{\mu} + \boldsymbol{\lambda}^T (\mathbf{A}^T \mathbf{n} - \mathbf{b}) + \eta (\mathbf{1}^T \mathbf{n} - n_T)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^C$ is the vector of Lagrange multipliers associated with the elemental balances, and $\eta \in \mathbb{R}$ is the multiplier associated with the total-mole constraint.

Expanding the chemical potentials, we obtain

$$L(\mathbf{n}, n_T, \boldsymbol{\lambda}, \eta) = \sum_{i=1}^{N} n_i \left[\mu_i^\circ + RT \ln\left(x_i \gamma_i\right)\right] + \sum_{j=1}^{C} \lambda_j \left(\sum_{i=1}^{N} A_{ij} n_i - b_j\right) + \eta(\mathbf{1}^T \mathbf{n} - n_T)$$

Dividing the entire expression by RT yields a dimensionless Lagrangian,

$$l = \frac{L}{RT},$$

which is advantageous numerically because it keeps the scale of the variables and equations under better control.

Define the dimensionless chemical potential vector by

$$\boldsymbol{\nu} = \frac{\boldsymbol{\mu}}{RT}$$

so that, componentwise,

$$\nu_i = \frac{\mu_i^\circ}{RT} + \ln(x_i \gamma_i).$$

We also absorb the factor $1/RT$ into the Lagrange multipliers, so that $\boldsymbol{\lambda}$ and η are from now on understood to be dimensionless. With this convention, the scaled Lagrangian becomes

$$l(\mathbf{n}, n_T, \boldsymbol{\lambda}, \eta) = \mathbf{n}^T \boldsymbol{\nu} + \boldsymbol{\lambda}^T (\mathbf{A}^T \mathbf{n} - \mathbf{b}) + \eta(\mathbf{1}^T \mathbf{n} - n_T).$$

To solve the resulting first-order optimality system with Newton-Raphson, we require the analytical gradient of this expression and the Jacobian of the corresponding KKT residual equations. Evaluating these quantities numerically is strongly discouraged, since small floating-point errors may severely degrade convergence.

3. Calculating the gradient

By definition, the differential of the scalar function l is the linear form that maps the differentials of its variables — namely $d\mathbf{n}$, dn_T, $d\boldsymbol{\lambda}$, and $d\eta$ — to the differential dl. Expanding dl, we obtain

$$dl = d\mathbf{n}^T \boldsymbol{\nu} + \mathbf{n}^T d\boldsymbol{\nu} + d\boldsymbol{\lambda}^T (\mathbf{A}^T \mathbf{n} - \mathbf{b}) + \boldsymbol{\lambda}^T \mathbf{A}^T d\mathbf{n} + d\eta(\mathbf{1}^T \mathbf{n} - n_T) + \eta(\mathbf{1}^T d\mathbf{n} - dn_T).$$

The most delicate term is $\mathbf{n}^T d\boldsymbol{\nu}$. To evaluate it, we first rewrite the dimensionless chemical potential vector using the identity

$$\ln(x_i) = \ln(n_i) - \ln(n_T).$$

This gives

$$\boldsymbol{\nu} = \boldsymbol{\nu}^\circ + \ln(\mathbf{n}) - \ln(n_T)\mathbf{1} + \ln(\boldsymbol{\gamma}),$$

where $\ln(\mathbf{n})$ and $\ln(\boldsymbol{\gamma})$ are understood componentwise.

Assuming that the activity coefficients are held fixed during the inner Newton-Raphson loop, we have

$$d\boldsymbol{\nu} = \operatorname{diag}(\mathbf{1} \oslash \mathbf{n})\, d\mathbf{n} - \frac{1}{n_T}\mathbf{1}\, dn_T,$$

where $\oslash$ denotes elementwise division.

Multiplying on the left by $\mathbf{n}^T$ yields

$$\mathbf{n}^T d\boldsymbol{\nu} = \mathbf{n}^T \operatorname{diag}(\mathbf{1} \oslash \mathbf{n})\, d\mathbf{n} - \mathbf{n}^T \mathbf{1} \frac{1}{n_T} dn_T.$$

Since

$$\mathbf{n}^T \operatorname{diag}(\mathbf{1} \oslash \mathbf{n}) = \mathbf{1}^T \qquad \text{and} \qquad \mathbf{n}^T \mathbf{1} = \sum_{i=1}^{N} n_i = \mathbf{1}^T \mathbf{n},$$

it follows that

$$\mathbf{n}^T d\boldsymbol{\nu} = \mathbf{1}^T d\mathbf{n} - \frac{\mathbf{1}^T \mathbf{n}}{n_T} dn_T.$$

Substituting this result back into the expression for dl and grouping terms by differential gives

$$\begin{aligned} dl = {} & d\mathbf{n}^T \left(\boldsymbol{\nu} + \mathbf{1} + \mathbf{A}\boldsymbol{\lambda} + \eta \mathbf{1}\right) \\ & + dn_T \left(-\frac{\mathbf{1}^T \mathbf{n}}{n_T} - \eta\right) \\ & + d\boldsymbol{\lambda}^T (\mathbf{A}^T \mathbf{n} - \mathbf{b}) \\ & + d\eta (\mathbf{1}^T \mathbf{n} - n_T). \end{aligned}$$

Therefore, the analytical gradient of the scaled Lagrangian is

$$\nabla l = \begin{bmatrix} \boldsymbol{\nu} + \mathbf{1} + \mathbf{A}\boldsymbol{\lambda} + \eta \mathbf{1} \\ -\dfrac{\mathbf{1}^T \mathbf{n}}{n_T} - \eta \\ \mathbf{A}^T \mathbf{n} - \mathbf{b} \\ \mathbf{1}^T \mathbf{n} - n_T \end{bmatrix}.$$

4. Calculating the Jacobian

The Newton-Raphson step is applied not directly to the scalar Lagrangian, but to the nonlinear first-order optimality system defined by the residual vector ∇l. We therefore require the Jacobian of this residual.

Differentiating the gradient, we obtain

$$d(\nabla l) = \begin{bmatrix} d\boldsymbol{\nu} + \mathbf{A}\, d\boldsymbol{\lambda} + \mathbf{1}\, d\eta \\ d\left(-\dfrac{\mathbf{1}^T \mathbf{n}}{n_T} - \eta\right) \\ \mathbf{A}^T d\mathbf{n} \\ \mathbf{1}^T d\mathbf{n} - dn_T \end{bmatrix}.$$

We have already shown that

$$d\boldsymbol{\nu} = \operatorname{diag}(\mathbf{1} \oslash \mathbf{n})\, d\mathbf{n} - \frac{1}{n_T}\mathbf{1}\, dn_T.$$

For the second block, applying the quotient rule gives

$$d\left(-\frac{\mathbf{1}^T\mathbf{n}}{n_T} - \eta\right) = -\frac{1}{n_T}\mathbf{1}^T d\mathbf{n} + \frac{\mathbf{1}^T\mathbf{n}}{n_T^2} dn_T - d\eta.$$

Collecting these relations into block-matrix form yields

$$d(\nabla l) = \begin{bmatrix} \operatorname{diag}(\mathbf{1} \oslash \mathbf{n}) & -\frac{1}{n_T}\mathbf{1} & \mathbf{A} & \mathbf{1} \\ -\frac{1}{n_T}\mathbf{1}^T & \frac{\mathbf{1}^T\mathbf{n}}{n_T^2} & \mathbf{0} & -1 \\ \mathbf{A}^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1}^T & -1 & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} d\mathbf{n} \\ dn_T \\ d\boldsymbol{\lambda} \\ d\eta \end{bmatrix}.$$

Hence, the exact Jacobian of the KKT residual system is

$$\mathbf{J}_l = \begin{bmatrix} \operatorname{diag}(\mathbf{1} \oslash \mathbf{n}) & -\frac{1}{n_T}\mathbf{1} & \mathbf{A} & \mathbf{1} \\ -\frac{1}{n_T}\mathbf{1}^T & \frac{\mathbf{1}^T\mathbf{n}}{n_T^2} & \mathbf{0} & -1 \\ \mathbf{A}^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1}^T & -1 & \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

This is the symmetric block matrix used in the Newton-Raphson step for the constrained minimization problem under the assumption of fixed activity coefficients during each inner iteration.

Practical note on convergence

The Newton-Raphson procedure derived here is fundamentally a local method. In many practical cases, it performs well when initialized near the physically relevant solution. Nevertheless, in strongly nonideal systems, the Gibbs energy surface may exhibit multiple local minima, and convergence to the global equilibrium state is then no longer guaranteed.

Accordingly, when global optimality is essential, one should consider global optimization procedures in addition to local Newton-type iterations; see, for example, [1].

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The course is freely available online through MIT OpenCourseWare and YouTube:

- MIT OpenCourseWare

- YouTube playlist

References

[1] MR Vianna Neto and ED Oliveira. Chemical and phase equilibrium calculations by gibbs energy minimization using deterministic methods based on globally convergent branch and bound algorithms. *Fluid Phase Equilibria*, 447:95–106, 2017.